

BME680

Temperature, Humidity, Pressure Sensor

Datasheet

The BME680 is a compact 4-in-1 environmental sensor from Bosch Sensortec that combines temperature, humidity, barometric pressure, and gas sensing for indoor air quality measurement. It supports both I²C and SPI interfaces, making it easy to integrate into a wide range of systems.

Designed to operate from $-40\text{ }^{\circ}\text{C}$ to $+85\text{ }^{\circ}\text{C}$ and across a pressure range of 300 to 1100 hPa, it delivers reliable performance in diverse environments. With its low power consumption and high accuracy, it is well suited for battery-powered devices. The BME680 is widely used in IoT applications, smart home devices, wearables, and mobile products for environmental monitoring and air quality tracking.

Key Features

- Integrated 4-in-1 environmental sensor (gas, pressure, humidity, temperature)
- Provides direct Indoor Air Quality (IAQ) index output
- Supports I²C and SPI digital interfaces
- Designed for low power consumption, suitable for battery devices
- Compact $3.0 \times 3.0 \times 0.93\text{ mm}$ package for easy integration



BME680 Sensor Board

Technical Specifications

Parameter	Value
Package	3.0 mm × 3.0 mm × 0.93 mm, metal lid LGA
Digital Interfaces	I ² C (up to 3.4 MHz); SPI, 3-/4-wire (up to 10 MHz)
Supply Voltage	VDD (main): 1.71 V to 3.6 V; VDDIO (interface): 1.2 V to 3.6 V
Current Consumption (typical)	2.1 μA @ 1 Hz (humidity + temperature); 3.1 μA @ 1 Hz (pressure + temperature); 3.7 μA @ 1 Hz (humidity + pressure + temperature); 0.09–12 mA depending on mode (including gas measurement); 0.15 μA in sleep mode
Operating Range	Temperature: $-40\text{ }^{\circ}\text{C}$ to $+85\text{ }^{\circ}\text{C}$; Humidity: 0 % to 100 % r.H.; Pressure: 300 hPa to 1100 hPa
Gas Sensor Parameters	Response time < 1 s (new sensors); power consumption < 0.1 mA in ultra-low power mode; output is a direct Indoor Air Quality (IAQ) index
Humidity Sensor Parameters	Response time ~8 s; accuracy tolerance $\pm 3\text{ }\%$ r.H.; hysteresis $\pm 1.5\text{ }\%$ r.H.
Pressure Sensor Parameters	RMS noise 0.12 Pa ($\approx 1.7\text{ cm}$); offset temperature coefficient $\pm 1.3\text{ Pa/K}$ ($\approx \pm 10.9\text{ cm per }1\text{ }^{\circ}\text{C}$)

Applications

- Indoor air quality
- Home automation and control, Internet of Things
- Weather forecasting
- GPS enhancement, etc.